

Microsoft® Official Course



Module 6

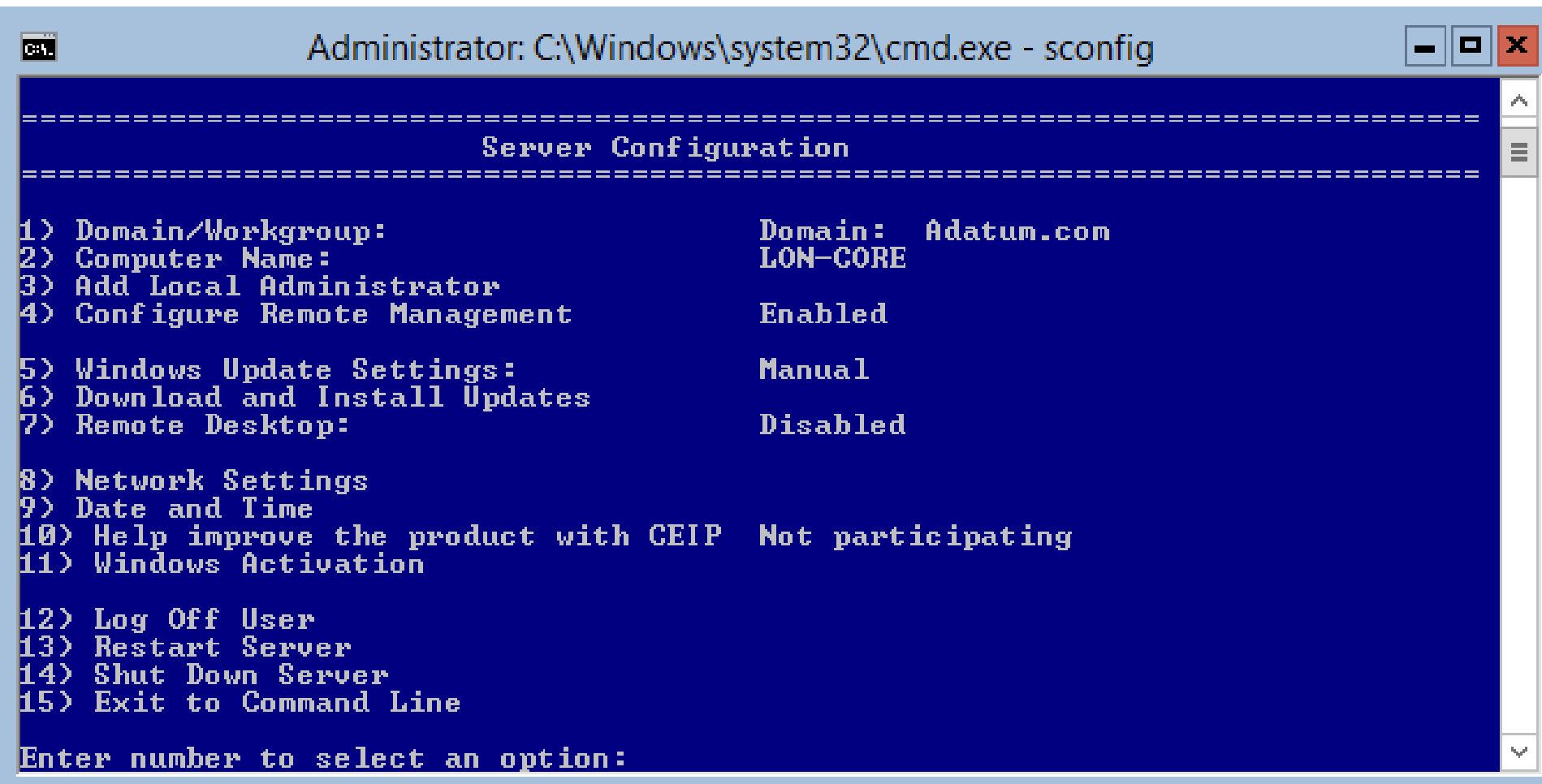
Using Windows PowerShell for Windows Server Administration

Server Core ?..

Server Core:

- Version de Windows Server 2012 plus sécurisée et plus légère ;
- Possibilité de passer de la version « Server Core » à la version standard (installation de l'interface graphique) ;
- Version par défaut de Windows Server 2012 ;
- Pour une gestion locale : « sconfig.cmd » ;
- Pour une gestion à distance : « remote management ».

Configuration d'un Server Core



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - sconfig". The console displays the "Server Configuration" menu with 15 numbered options. The current configuration is as follows:

Option	Configuration
1) Domain/Workgroup:	Domain: Adatum.com
2) Computer Name:	LON-CORE
3) Add Local Administrator	
4) Configure Remote Management	Enabled
5) Windows Update Settings:	Manual
6) Download and Install Updates	
7) Remote Desktop:	Disabled
8) Network Settings	
9) Date and Time	
10) Help improve the product with CEIP	Not participating
11) Windows Activation	
12) Log Off User	
13) Restart Server	
14) Shut Down Server	
15) Exit to Command Line	

Enter number to select an option:

Windows PowerShell

Administrator: Windows PowerShell

Name	ModuleName
----	-----
Set-BCAuthentication	BranchCache
Set-BCCache	BranchCache
Set-BCDataCacheEntryMaxAge	BranchCache
Set-BCMinSMBLatency	BranchCache
Set-BCSecretKey	BranchCache
Set-ClusteredScheduledTask	ScheduledTasks
Set-DAClientExperienceConfiguration	DirectAccessClientComponents
Set-DAEntryPointTableItem	DirectAccessClientComponents
Set-Disk	Storage
Set-DnsClient	DnsClient
Set-DnsClientGlobalSetting	DnsClient
Set-DnsClientNrptGlobal	DnsClient
Set-DnsClientNrptRule	DnsClient
Set-DnsClientServerAddress	DnsClient
Set-DtcAdvancedHostSetting	MsDtc
Set-DtcAdvancedSetting	MsDtc
Set-DtcClusterDefault	MsDtc
Set-DtcClusterTMMapping	MsDtc
Set-DtcDefault	MsDtc
Set-DtcLog	MsDtc
Set-DtcNetworkSetting	MsDtc
Set-DtcTransaction	MsDtc
Set-DtcTransactionsTraceSession	MsDtc
Set-DtcTransactionsTraceSetting	MsDtc
Set-InitiatorPort	Storage
Set-iSCSIChapSecret	iSCSI
Set-LogProperties	PSDiagnostics
Set-MMAgent	MMAgent
Set-NCSIPolicyConfiguration	NetworkConnectivityStatus
Set-Net6to4Configuration	NetworkTransition
Set-NetAdapter	NetAdapter
Set-NetAdapterAdvancedProperty	NetAdapter
Set-NetAdapterBinding	NetAdapter
Set-NetAdapterChecksumOffload	NetAdapter

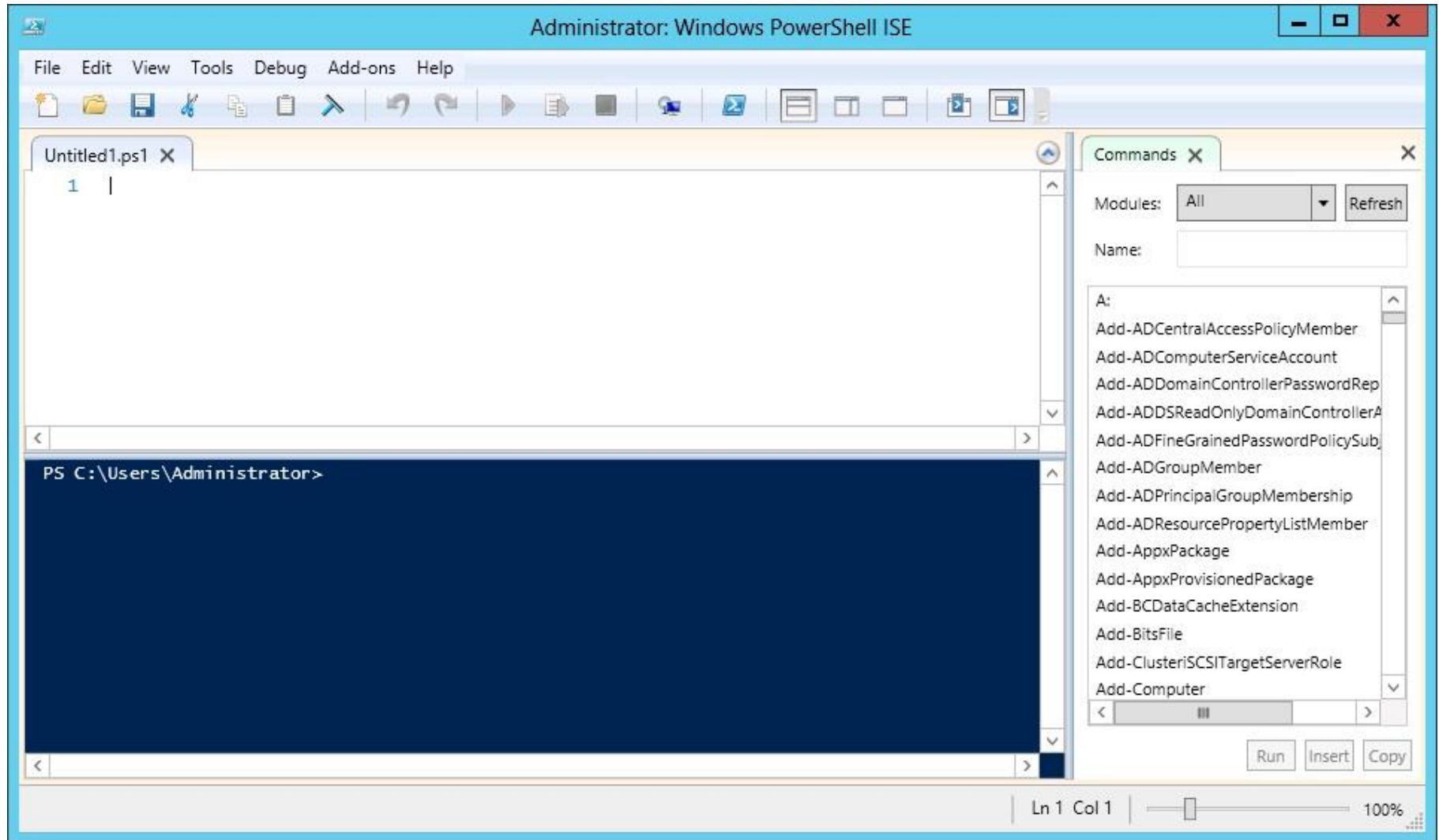
Windows PowerShell : Syntaxe

Quelques astuces :

- `Get-Help NounName`
- `Get-Help VerbName`
- `Help CmdltName`

Liste des principaux verbes/actions : *Get, New, Set, Restart, Resume, Stop, Suspend, Clear, Limit, Remove, Add, Show, Write, ...*

Windows PowerShell ISE



Déploiement du rôle AD DS

- En utilisant les commandes power shell (option Core) :
 - Activation du rôle
Install-WindowsFeature AD-Domain-Services
 - Ajouter un contrôleur de domaine supplémentaire
Install-ADDSDomainController
 - Ajouter un domaine supplémentaire
Install-ADDSDomain
 - Créer un nouveau domaine
Install-ADDSTForest

Installation d'un contrôleur de domaine

En utilisant les commandes power shell (Core) :

1. Installation des fichiers du rôle

Install-WindowsFeature AD-Domain-Services

2. Création et configuration du contrôleur de domaine

Install-ADDSDomainController

Using Windows PowerShell for AD DS Administration

- Using Windows PowerShell Cmdlets to Manage User Accounts
- Using Windows PowerShell Cmdlets to Manage Groups
- Using Windows PowerShell Cmdlets to Manage OUs
- Querying Objects with Windows PowerShell
- Modifying Objects with Windows PowerShell

Using Windows PowerShell Cmdlets to Manage User Accounts

Cmdlet	Description
New-ADUser	Creates user accounts
Set-ADUser	Modifies properties of user accounts
Remove-ADUser	Deletes user accounts
Set-ADAccountPassword	Resets the password of a user account
Set-ADAccountExpiration	Modifies the expiration date of a user account
Unlock-ADAccount	Unlocks a user account after it has become locked after too many incorrect login attempts
Enable-ADAccount	Enables a user account
Disable-ADAccount	Disables a user account

```
New-ADUser "Sten Faerch" -AccountPassword (Read-Host  
-AsSecureString "Enter password") -Department IT
```

Création d'un compte utilisateur

En utilisant Windows PowerShell

```
New-ADUser -Name "Dan Park" -SamAccountName  
  "Dan" -Path "OU=Sales,DC=Adatum,DC=com"  
  -AccountPassword (ConvertTo-SecureString -  
    AsPlaintext "Pa$$w0rd" -Force)  
-GivenName "Dan" -Surname "Park" -DisplayName  
  "Dan Park" -Enabled $True  
  -UserPrincipalName "Dan@Adatum.com" -  
    ChangePasswordAtLogon $false
```

Gestion de Comptes Utilisateur (PowerShell)

- Changement de mot de passe :
 - **Set-ADAccountPassword** -identity 'cn=amy, ou=IT, dc=contoso, dc=com' -Reset -NewPassword (ConvertTo SecureString -AsPlainText "Pa\$\$w0rd2" -Force)
- Déverrouiller un compte utilisateur
 - **Unlock-ADAccount** -identity 'cn=amy strand, ou=IT, dc=contoso, dc=com'
- Activer/Désactiver un compte utilisateur
 - **Enable-ADAccount** -identity <name>
 - **Disable-ADAccount** -identity <name>

Using Windows PowerShell Cmdlets to Manage Groups

Cmdlet	Description
New-ADGroup	Creates new groups
Set-ADGroup	Modifies properties of groups
Get-ADGroup	Displays properties of groups
Remove-ADGroup	Deletes groups
Add-ADGroupMember	Adds members to groups
Get-ADGroupMember	Displays membership of groups
Remove-ADGroupMember	Removes members from groups
Add-ADPrincipalGroupMembership	Adds group membership to objects
Get-ADPrincipalGroupMembership	Displays group membership of objects
Remove-ADPrincipalGroupMembership	Removes group membership from an object

```
New-ADGroup -Name "CustomerManagement" -Path  
"ou=managers,dc=adatum,dc=com" -GroupScope Global  
-GroupCategory Security
```

```
Add-ADGroupMember -Name "CustomerManagement"  
-Members "Joe"
```

Using Windows PowerShell Cmdlets to Manage OUs

Cmdlet	Description
New-ADOrganizationalUnit	Creates OUs
Set-ADOrganizationalUnit	Modifies properties of OUs
Get-ADOrganizationalUnit	Views properties of OUs
Remove-ADOrganizationalUnit	Deletes OUs

```
New-ADOrganizationalUnit -Name "Sales"  
-Path "ou=marketing,dc=adatum,dc=com"  
-ProtectedFromAccidentalDeletion $true
```

Querying Objects with Windows PowerShell

Parameter	Description
SearchBase	Defines the AD DS path to begin searching
SearchScope	Defines at what level below the SearchBase a search should be performed
ResultSetSize	Defines how many objects to return in response to a query
Properties	Defines which object properties to return and display
Filter	Defines a filter by using PowerShell syntax
LDAPFilter	Defines a filter by using LDAP query syntax

Descriptions of operators

-eq	Equal to	-gt	Greater than
-ne	Not equal to	-ge	Greater than or equal to
-lt	Less than	-like	Uses wildcards for pattern matching
-le	Less than or equal to		



Querying Objects with Windows PowerShell

Show all the properties for a user account:

```
Get-ADUser -Name "Administrator" -Properties *
```

Show all the user accounts in the Marketing OU and all its subcontainers:

```
Get-ADUser -Filter * -SearchBase  
"ou=Marketing,dc=adatum,dc=com" -SearchScope subtree
```

Show all of the user accounts with a last logon date older than a specific date:

```
Get-ADUser -Filter {lastlogondate -lt "January 1, 2012"}
```

Show all of the user accounts in the Marketing department that have a last logon date older than a specific date:

```
Get-ADUser -Filter {(lastlogondate -lt "January 1,  
2012")} -and (department -eq "Marketing")}
```



Modifying Objects with Windows PowerShell

Use the pipe character (|) to pass a list of objects to a cmdlet for further processing

```
Get-ADUser -Filter {company -notlike "*"} |  
Set-ADUser -Company "A. Datum"
```

```
Get-ADUser -Filter {lastlogondate -lt "January 1,  
2012"} | Disable-ADAccount
```

```
Get-Content C:\users.txt | Disable-ADAccount
```

Exercice

- Créez un compte utilisateur avec les propriétés suivantes : nom=Paul Dupont, SamAN=pdupont, UPN=pdupont@adatum.com, département=RH ;
- Créez 3 autres utilisateurs travaillant également dans le département RH ;
- Affichez tous les utilisateurs du département RH ;
- Créez un groupe global RH ;
- Créez une unité d'organisation Administration ;
- Ajoutez les utilisateurs et le groupe au sein de l'OU ;
- Affichez tous les objets de l'OU ;
- Ajoutez les utilisateurs au sein du groupe ;
- Affichez tous les utilisateurs du groupe ;
- Activez tous les comptes ;